| Name of param | Default value | Involved functions | Description | Unit | Category | Datatype |
|---|---|---|---|---|---|---|
| temperature | 20,338 | process_data | Temperature in celsius. | celsius | env_params | float |
| salinity | 34,27 | process_data | Salinity from PPS78 (R gsw_SP_from_C) | PSU | env_params | float |
| pressure | 10,1325 | process_data | Pressure in dBar. | dBar | env_params | float |
| gain_correction | 28,49 | process_data | Center value of gain. | | cal_params | float |
| equivalent_beam_angle | −21 | process_data | Ctrl+F in the calibration file to find val. | | cal_params | float |
| bin_size | 0,1 | process_data | Bin size to average along range in meters (y-axis). | meters | | float |
| new_file_path | ? | data_to_images, main.py | Filepath to created file. Not sure if this will be used onboard. | | | string |
| layer_strength_thresh | −80 | find_layer | Threshold to find layers at the top of the files. Signals stronger than threshhold will be considered part of layer. | dB | | float |
| layer_range_in_a_row | 3 | find_layer | Number of signals in a row in one ping that has to be False *(smaller than layer_strength_thresh)* to stop the find_layer algorithm. | | | int |
| bottom_offset | 20 | find_bottom, find_fish_median | Number of pixles to remove at the bottom due to the echo effect from seabeds deeper than 100m. | | | int |
| bottom_hardness_thresh | −20 | find_bottom | Threshold used to classify bottom. Signal needs to be stronger than threshold to be classified as bottom. If signal is weaker, depth > 100m. | dB | | float |
| bottom_roughness_thresh | 30 | find_bottom | The mean roughness (the difference in number of bins between the detected bottom in each ping) is not allowed to be greater than this thresh if a bottom is classified in the data file. | | | float |
| no_bottom_default | 101 | find_bottom | Default bottom value if no bottom is found. | meters | | float |
| beam_dead_zone | 15 | find_waves, main.py, find_fish_median, find_layer | Size of echosounder near-zone, number of bins to remove at the top of the echodata. | | | int |
| beam_dead_zone_offset | *beam_dead_zone* + 5 | find_waves | Number of extra pixles to remove at top. (Not sure if necessary, remove?) | | | int |
| layer_quantile | 0,7 | find_layer | Which quantile to use when looking at each row to find average echo strength. | | | float (between 0 and 1) |
| wave_thresh | −77 | find_waves | Threshold to use when looking for waves. Echo needs to be stronger than this threshold to be classified as part of wave. | dB | | float |
| wave_thresh_layer | −68 | find_waves | Threshold to use when looking for waves when a layer is existing in the echodata. Echo needs to be stronger than this threshold to be classified as part of wave. | dB | | float |
| extreme_wave_size | 70 | main.py, find_waves(move here?) | Average wave size limit to consider detected wave as unrealistic big in echodata. | | | float |
| in_a_row_waves | 3 | find_waves | Number of signals in a row that has to be false (<wave_thresh) to consider wave as finished. | | | int |
| move_avg_windowsize | 9 | move_fun | Windowsize to use when calculating moving averages. | | | int |
| fish_layer0_start | 0 | median_fun | Where to start measuring fish layer0, this layer should include total. | meters | | int |
| fish_layer0_end | 100 | median_fun | Where to stop measuring fish layer0, this layer should include total. | meters | | int |
| fish_layer1_start | 0 | median_fun | Where to start measuring fish layer1. | meters | | int |
| fish_layer1_end | 35 | median_fun | Where to stop measuring fish layer1. | meters | | int |
| fish_layer2_start | 35 | median_fun | Where to start measuring fish layer2. | meters | | int |

| | | | | | |
|---|---|---|---|---|---|
| fish_layer2_end | 80 | median_fun | Where to stop measuring fish layer2. | meters | int |
| fish_layer3_start | 80 | median_fun | Where to start measuring fish layer3. | meters | int |
| fish_layer3_end | 100 | median_fun | Where to stop measuring fish layer3. | meters | int |
| save_fig | FALSE | activation of data_to_images in main.py | If True: save images and csv-file from each run. If False: don't save anything. Not sure if this will be used onboard. | | bool |

| Name of output | Description | Description of calculations | Involved functions | Unit | other remarks |
|---|---|---|---|---|---|
| time | The time of the first ping | - From **process_data** a list of times for every ping is extracted from the raw data<br>- If rows with all NaN values are found in **remove_vertical_lines** corresponding times are removed in **clean_times**<br>- At summarization the first time is chosen | process_data, remove_vertical_lines, clean_times | YYYY-MM-DD HH:MM:SS (UTC) | |
| bottom_depth | The median bottom depth | - **Maxecho** gets the depth of highest value for every ping excluding the first and last 20 rows (20 dm of depth)<br>- The list gets smoothed with moving averages in **move_fun**<br>- All depths gets subtracted by 20<br>- All depths gets multiplied by 0.1 to get meters.<br>- Summarization: median from depths list | find_bottom, maxecho, move_fun | m | - Lägg till 'dead zone' 53 cm 'sonar depth' som parameter |
| bottom_hardness | The median bottom echostrength | - **maxval** gets the higest value for every ping<br>- Summarization: median from list | find_bottom, maxval | db | |
| bottom_roughness | The mean bottom depth change | - Calculates the average change between consecutive depths<br>- Rounds the result to two decimal places. | find_bottom | dm | - Ändra till meter |
| wave_depth | The median wave depth | - Removes the first 15 rows<br>- Limit is chosen<br>- Searches for 3 consecutive values below limit to signal break of wave. Appends index of depth subtracted by 5 to wave_depth list.<br>- If not found the total depth of ping subtracted by 2 gets appended to wave_depth list<br>- Every depth in wave_depth list gets added by 16 | find_waves | dm | - Ändra till meter<br>- Lägg till surf offset och sonar depth som parameter<br>- Ta bort onödig datas list<br>**-** Ändra det till mean |

| | | | | | |
|---|---|---|---|---|---|
| nasc0 | The median nasc per ping | - In **find_fish_median2** the nasc of every datapoint is calculated. Excluding 5 dm under wave line and 16 dm over bottom<br>- In medianfun2 a list of cumalative sum for every ping is calculated<br>- Summarization: median from list | find_fish_median2, medianfun2 | m2 nmi 2 | |
| fishdepth0 | The median depth of fish per ping | - The array from find_fish_median2 is used to calculate the median depth of fish for every ping<br>- List gets divided by 10 and converted to meters<br>- Summarization: median from list | find_fish_median2, medianfun2 | m | - lägg till sonar depth |
| nasc1 | The median nasc per ping from, depth 0-35 | Same calculations as nasc0 but pings get sliced beforehand | find_fish_median2, medianfun2 | ? | |
| fishdepth1 | The median depth of fish per ping, depth 0-35 | | find_fish_median2, medianfun2 | m | lägg till sonar depth |
| nasc2 | The median nasc per ping from, depth 35-80 | | find_fish_median2, medianfun2 | ? | |
| fishdepth2 | The median depth of fish per ping, depth 35-80 | | find_fish_median2, medianfun2 | m | lägg till sonar depth |
| nasc3 | The median nasc per ping from, depth 80-100 | | find_fish_median2, medianfun2 | ? | |
| fishdepth3 | The median depth of fish per ping, depth 80-100 | | find_fish_median2, medianfun2 | m | lägg till sonar depth |